Question	Answer	Marks
3(a)	time = 12 s	**A1**
3(b)	distance (up slope) = ½ × 12 × 18 (= 108)	**C1**
	distance (down slope) = ½ × 12 × 6 (= 36)	**C1**
	displacement from A = 108 – 36 = 72 m	**A1**
3(c)	$v = u + at$ **or** a = gradient **or** $a = \Delta v / (\Delta)t$	**C1**
	$a = 6 / 12 = 0.50$ (m s^{-2}) *(other points from the line may be used)*	**A1**
	or	
	$v^2 = u^2 + 2as$ **and** $u = 0$ **or** $v^2 = 2as$	**(C1)**
	$a = 6.0^2 / (2 \times 36) = 0.50$ (m s^{-2})	**(A1)**
	or	
	$s = ut + ½at^2$ **and** $u = 0$ **or** $s = ½at^2$	**(C1)**
	$a = 2 \times 36 / 12^2 = 0.50$ (m s^{-2})	**(A1)**
	or	
	$s = vt – ½at^2$	**(C1)**
	$a = 2 \times (6 \times 12 – 36) / 12^2 = 0.50$ (m s^{-2})	**(A1)**

Question	Answer	Marks
3(d)(i)	$F = 70 \times 0.50$ (= 35)	**C1**
	frictional force = 80 – 35 = 45 N	**A1**
3(d)(ii)	$\sin \theta = 80 / (70 \times 9.81)$	**C1**
	$\theta = 6.7°$	**A1**
3(e)(i)	$f_0 = (900 \times 340) / (340 + 12)$	**C1**
	= 870 Hz	**A1**
3(e)(ii)	speed/velocity (of sledge) decreases and (so) frequency increases	**B1**

Question	Answer	Marks
4(a)(i)	distance moved by wavefront/energy during one cycle/oscillation/period (of source)	B1